E-Commerce Product & Sales Data Analysis

1. Project Overview:

- **Objective:** This project analyzes e-commerce order and transactional data to uncover actionable insights into sales performance, product profitability, regional demand, and order fulfillment trends.
- **Scope:** An end-to-end data analytics workflow covering data cleaning and feature engineering (Python), relational database querying (SQL), interactive dashboard creation (Power BI), and strategic business recommendations.

2. Dataset Summary:

- **Dimensions:** 380 cleaned order records and 21 columns (refined from 524 raw records and 15 columns).
- **Key Attributes:** Order ID, Order Date, Customer Name, City, State, Product, Category, Qty, Unit Price, Discount, Payment Mode, Order Status, Delivery Date, Sales, Net Amount, Profit, Month, Year, and Weekday.
- **Data Quality:** Removed duplicate records, filtered invalid quantities, standardized text formats, and imputed missing values.

- **Statistical Highlights:**
    - **Total Sales (Net Revenue):** ₹35,15,432.15 (**3.52M**) *(Gross Sales: ₹38,08,557.00 / 3.81M)*
    - **Total Orders: 374** valid Order IDs *(out of 380 cleaned rows)*
    - **Total Net Profit (20% Margin):** ₹7,03,086.43 (**703.09K**)
    - **Average Order Value (Gross):** ₹9,399.55 (**9.40K**)

3. Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python (Pandas and NumPy):

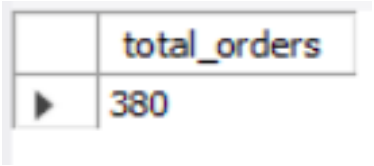
- **Data Loading & Inspection:** Imported the raw Excel dataset using pandas and verified structure and summary statistics via data.info() and data.describe().
- **Data Cleaning:** Removed 11 duplicate rows, replaced invalid placeholders (N/A, NULL, blanks) with NaN, filtered out negative/zero quantities, and validated email addresses.
- **Missing Data Handling:** Imputed missing Discount values with 0, missing Phone numbers with "Unknown", and missing Delivery_Date entries with Order_Date.
- **Feature Engineering:**

    - Created Sales, Net_Amount, and Profit columns to evaluate post-discount revenue and margins.
    - Extracted Month, Year, and Weekday from Order_Date for time-series analysis.
- **Database Integration:** Connected Python to MySQL (ecommerce database) using SQLAlchemy and exported the cleaned DataFrame into the orders table for SQL analysis.

4. Data Analysis using SQL:

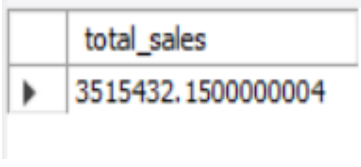
We performed structured analysis in MySQL to answer key business questions:

1. **Total Orders** – Calculated the total count of valid orders recorded in the database (380 orders).



	total_orders
▶	380

2. **Total Sales** – Computed total net revenue generated after discounts (₹35,15,432.15).



	total_sales
▶	3515432.1500000004

3. **Top Product by Sales** – Ranked products by net sales, identifying **Laptop** (₹6,77,183) and **Chair** (₹6,62,832.80) as top revenue generators.

	product	sales
▶	Laptop	677182.9999999999
	Chair	662832.8
	Mouse	653062.75
	Bottle	598722.1000000001
	Watch	498299.64999999997
	Shoes	425331.85

4. **Top Cities by Sales** – Compared city-wise net sales, led by **Pune** (₹8,25,418.60) and **Bengaluru** (₹7,66,916.95).

	city	sales
▶	Pune	802923.1000000001
	Bengaluru	731923.9500000001
	Lucknow	533284.3500000001
	Delhi	508417
	Mumbai	450417.44999999995
	Jaipur	386376.9
	Bengaluru	34993
	Mumbai	23910.5
	Pune	22495.5
	Delhi	17093
	Lucknow	1999
	Jaipur	1598.4

5. **Monthly Sales Trend** – Analyzed month-over-month sales, identifying **May** (₹7,09,917.05) as the highest-performing month and **March** (₹2,44,864.30) as the lowest.

	month	sales
▶	June	653698.35
	February	643201.3
	May	709917.0500000002
	NULL	349630.24999999994
	April	549461.25
	January	364659.64999999997
	March	244864.3

6. **Highest Profit Products** – Identified the most profitable products, led by **Laptop** (₹1,35,436.60) and **Chair** (₹1,32,566.56).

	product	profit
▶	Laptop	135436.6
	Chair	132566.56000000003
	Mouse	130612.55
	Bottle	119744.42000000003
	Watch	99659.93000000001
	Shoes	85066.36999999997

7. **Payment Mode Distribution** – Evaluated order volume across payment methods (**Wallet**: 87, **COD**: 86, **NetBanking**: 85, **UPI**: 68, **Card**: 54).

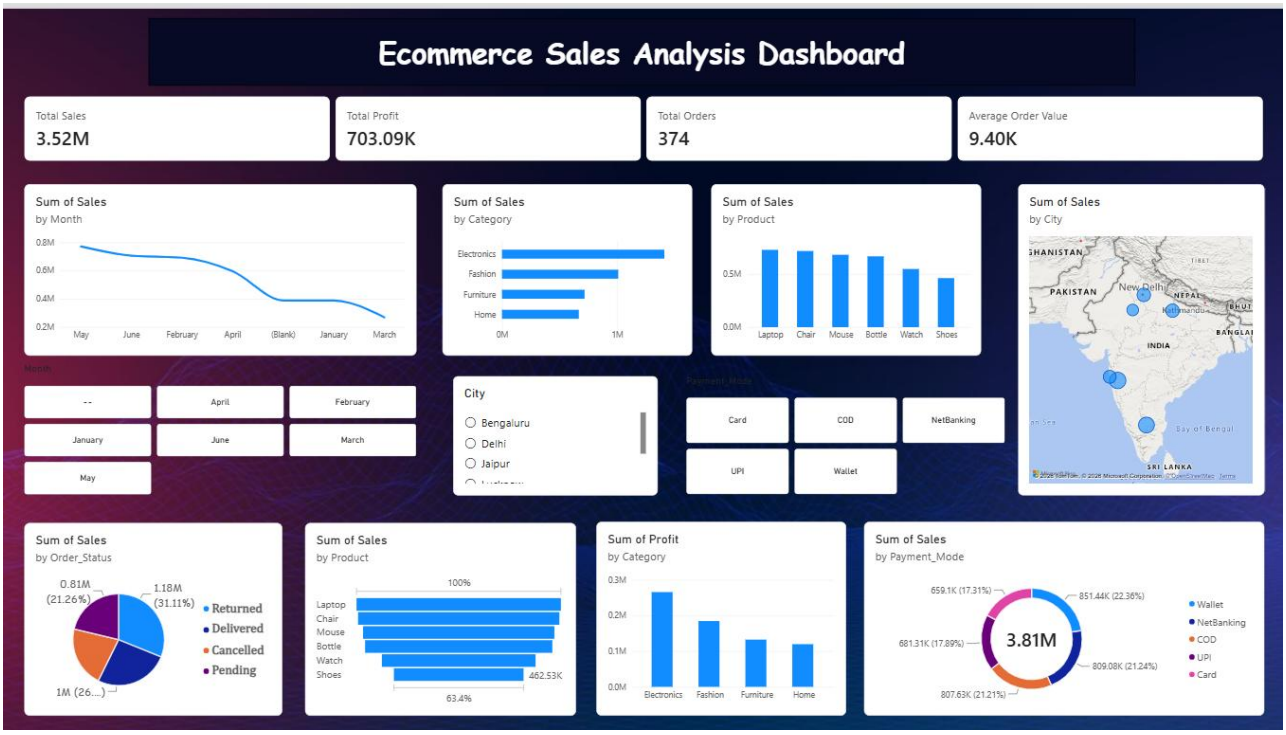
	payment_mode	total_orders
▶	UPI	68
	Card	54
	COD	86
	Wallet	87
	NetBanking	85

8. **Cancelled Orders** – Isolated cancelled transactions (94 orders) to track fulfillment bottlenecks and revenue loss.

	Order_ID	Order_Date	Customer_Name	Email	Phone	City	State	
▶	ORD1007	2025-02-10 00:00:00	Rahul	rahul82@gmail.com	9517780193	Mumbai	Maharashtra	M
	ORD1012	2025-03-03 00:00:00	Vikas	vikas46@gmail.com	9551158398	Jaipur	Rajasthan	S
	ORD1021	2025-01-27 00:00:00	Priya	priya82@gmail.com	9583849937	Pune	Maharashtra	C
	ORD1024	2025-06-29 00:00:00	Rohit	rohit60@gmail.com	9633404088	Bengaluru	Karnataka	M
	ORD1028	2025-06-05 00:00:00	Amit	amit15@gmail.com	9853264878	Lucknow	Up	W
	ORD1047	2025-03-17 00:00:00	Rohit	rohit24@gmail.com	9603990367	Jaipur	Rajasthan	S
	ORD1056	2025-05-29 00:00:00	Neha	neha98@gmail.com	9481676392	Pune	Maharashtra	C
	ORD1058	2025-02-12 00:00:00	Sneha	sneha17@gmail.com	9581041302	Jaipur	Rajasthan	C
	ORD1061	2025-06-01 00:00:00	Vikas	vikas12@gmail.com	9714988193	Jaipur	Rajasthan	S
	ORD1064	2025-06-28 00:00:00	Rohit	rohit44@gmail.com	9745435168	Lucknow	Up	B
	ORD1069	2025-01-20 00:00:00	Vikas	vikas55@gmail.com	9385463872	Lucknow	Up	S
	ORD1070	2025-02-28 00:00:00	Sneha	sneha18@gmail.com	9191665927	Delhi	Delhi	C
	NULL	2025-01-26 00:00:00	Vikas	vikas17@gmail.com	9814518863	Mumbai	Maharashtra	C
	ORD1077	2025-01-24 00:00:00	Neha	neha85@gmail.com	9919894416	Mumbai	Maharashtra	B
	ORD1083	2025-05-14 00:00:00	Priya	priya17@gmail.com	9678334449	Bengaluru	Karnataka	S
	ORD1088	2025-05-04 00:00:00	Rahul	rahul31@gmail.com	Unknown	Bengaluru	Karnataka	B
	ORD1095	2025-04-05 00:00:00	Rahul	rahul23@gmail.com	9141462098	Delhi	Delhi	C
	ORD1103	2025-01-26 00:00:00	Rohit	rohit37@gmail.com	9792751135	Pune	Maharashtra	M

5. Dashboard in Power BI:

Finally, we built an interactive dashboard in Power BI (Ecommerce Sales Dashboard.pbix) to present insights visually:



6. Business Recommendations:

- **Reduce Cancellations & Returns:** Audit order fulfillment and delivery timelines to minimize cancelled (94) and returned (97) orders and protect net revenue.
- **Focus on High-Performing Hubs:** Prioritize inventory and targeted marketing in top revenue cities like **Pune** and **Bengaluru** while expanding reach in emerging markets.

- **Boost Off-Peak Sales:** Introduce promotional bundles and flash deals during slower months like **March** to stabilize monthly revenue.
- **Incentivize Digital Payments:** Offer instant checkout discounts on **UPI, Wallet, and Card** payments to shift customers away from COD and reduce order cancellations.